

8th-Grade

TOPICS

1. Real Numbers
    - Reviews
    - Powers and Radicals
2. Algebraic Expressions
    - Definition of Algebraic Expressions
    - Identities and Factorization
    - Algebraic Ratios
3. Sequences and Functions
    - Sequences
    - Functions
    - Special Functions
4. Equations and Inequalities
    - Equations
    - System of Equations
    - Inequalities
5. Quadratic Equations
6. Probability and Statistics
    - Probability
    - Statistics
7. Mathematical Logic
8. Equations and Inequalities
    - Basic Geometric Concepts
    - Angle and Similarity in Triangles
    - Right Triangles and Trigonometry
    - Quadrilaterals

8th-Grade



2. Algebraic Expressions

Identities and Factorization

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.1 SQUARE OF BINOMIAL

Identity I

$$(a + b)^2 = a^2 + 2ab + b^2$$

EXAMPLE	SOLUTION
Expand $(x + 2)^2$.	$(x + 2)^2 = x^2 + 2 \cdot x \cdot 2 + 2^2 = x^2 + 4x + 4.$
Expand $(3x + 2y)^2$.	$(3x + 2y)^2 = (3x)^2 + 2 \cdot (3x) \cdot (2y) + (2y)^2 = 9x^2 + 12xy + 4y^2.$
Expand $(x + \frac{y}{2})^2$.	$(x + \frac{y}{2})^2 = x^2 + 2 \cdot x \cdot \frac{y}{2} + (\frac{y}{2})^2 = x^2 + xy + \frac{y^2}{4}.$

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.1 SQUARE OF BINOMIAL

Identity II

$$(a - b)^2 = a^2 - 2ab + b^2$$

EXAMPLE	SOLUTION
Expand $(x - 2)^2$.	$(x - 2)^2 = x^2 - 2 \cdot (x) \cdot (2) + 2^2 = x^2 - 4x + 4.$
Expand $(x^2 - 1)^2$.	$(x^2 - 1)^2 = (x^2)^2 - 2 \cdot (x^2) \cdot (1) + 1^2 = x^4 - 2x^2 + 1$
Expand $(2x - 3y)^2$.	$(2x - 3y)^2 = (2x)^2 - 2 \cdot (2x) \cdot (3y) + (3y)^2 = 4x^2 - 12xy + 9y^2$

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.1 SQUARE OF BINOMIAL

PRACTICE QUESTIONS

1. Expand the expressions.

a. $(x + 3)^2$

b. $(x + \frac{1}{2})^2$

c. $(\sqrt{x} + 2)^2$

d. $(-3x + 4)^2$

e. $(2x + y^2)^2$

f. $(xy + 3)^2$

g. $(x^2 + \frac{2}{3}y^2)^2$

h. $(\sqrt{x} + \sqrt{y})^2$

i. 53^3

j. 104^2

k. 97^2

l. 48^2

m. $(2x - 1)^2$

n. $(2x - 3y)^2$

o. $(3m - n)^2$

p. $(2xy - 3)^2$

q. $(p^2 - q^2)^2$

r. $(x - \frac{1}{2x})^2$

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

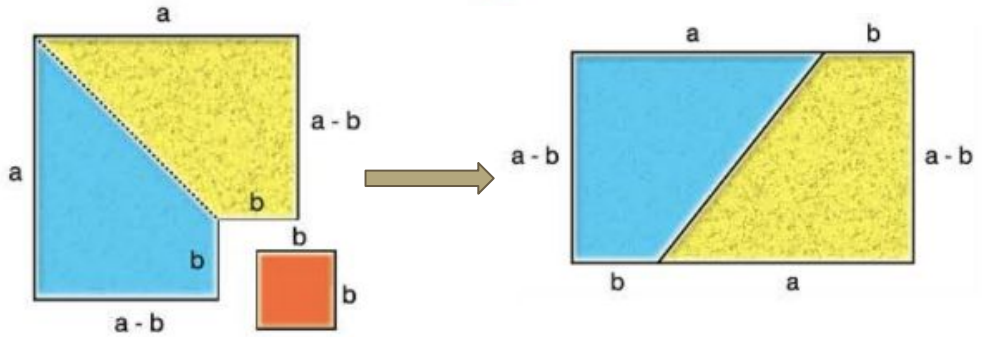
2.1.2 PRODUCT OF SUM AND DIFFERENCE OF 2 TERMS

Rule

$(a + b) \cdot (a - b) = a^2 - b^2$

$(a + b) \cdot (a - b) = a^2 - ab + ab - b^2 = a^2 - b^2.$

Proof:



The area of the rectangle is $(a - b) \cdot (a + b)$.

So $a^2 - b^2 = (a - b) \cdot (a + b)$.

8th-Grade

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.2 PRODUCT OF SUM AND DIFFERENCE OF 2 TERMS

Rule

$(a + b) \cdot (a - b) = a^2 - b^2$



$(a + b) \cdot (a - b) = a^2 - ab + ab - b^2 = a^2 - b^2.$

Example:

Find the products.

a. $(x + 4) \cdot (x - 4)$

b. $(\sqrt{x} + 1) \cdot (\sqrt{x} - 1)$

c. $(2a - b) \cdot (2a + b)$

d. $12 \cdot 8$

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.2 PRODUCT OF SUM AND DIFFERENCE OF 2 TERMS

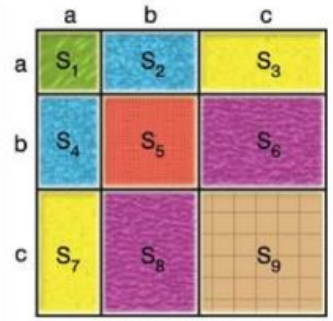
Rule

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$$

We can obtain the same result using geometry. Look at the rectangle.

$$S_1 = a^2$$
$$S_2 = a \cdot b$$
$$S_3 = a \cdot c$$
$$S_4 = b \cdot a$$
$$S_5 = b^2$$
$$S_6 = b \cdot c$$
$$S_7 = c \cdot a$$
$$S_8 = c \cdot b$$
$$+\ S_9 = c^2$$

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$$



8th-Grade

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.3 SQUARE OF TRINOMIALS

Rule

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$$

EXAMPLE

Expand $(2x - 3y - z)^2$.

SOLUTION

$$\begin{aligned}(2x - 3y - z)^2 &= (2x + (-3y) + (-z))^2 \\ &= (2x)^2 + (-3y)^2 + (-z)^2 + 2 \cdot (2x) \cdot (-3y) + 2 \cdot (2x) \cdot (-z) + 2 \cdot (-3y) \cdot (-z) \\ &= 4x^2 + 9y^2 + z^2 - 12xy - 4xz + 6yz\end{aligned}$$

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.3 SQUARE OF TRINOMIALS

Rule

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$$

EXAMPLE

Expand $(4a + b - 5c)^2$.

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.4 CUBE OF BINOMIALS

Rule

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

Proof 1

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$\begin{aligned}(a + b)^3 &= (a + b) \cdot (a + b) \cdot (a + b) \\ &= (a + b) \cdot (a^2 + 2ab + b^2) \\ &= a^3 + 2a^2 \cdot b + ab^2 + a^2 \cdot b + 2a \cdot b^2 + b^3 \\ &= a^3 + 3a^2b + 3ab^2 + b^3.\end{aligned}$$

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.4 CUBE OF BINOMIALS

EXAMPLE

Expand the expressions.

a. $(a + 1)^3$

b. $(x + 2)^3$

c. $(\frac{x}{2}+2y)^3$

SOLUTION

a. $(a + 1)^3 = a^3 + 3a^2 \cdot 1 + 3 \cdot a \cdot 1^2 + 1^3$

$= a^3 + 3a^2 + 3a + 1.$

b. $(x + 2)^3 = x^3 + 3x^2 \cdot 2 + 3 \cdot x \cdot 2^2 + 2^3$

$= x^3 + 3x^2 + 12x + 8.$

c. $(\frac{x}{2}+2y)^3 = (\frac{x}{2})^3 + 3 \cdot (\frac{x}{2})^2 \cdot (2y) + 3 \cdot (\frac{x}{2}) \cdot (2y)^2 + (2y)^3$

$= \frac{x^3}{8} + \frac{3}{2}x^2y + 6xy^2 + 8y^3.$

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.4 CUBE OF BINOMIALS

PRACTICE QUESTIONS

1. Expand the expressions.

a. $(x + 1)^3$

b. $(\frac{x}{2} + 3)^3$

c. $(x + 2y)^3$

d. $(2x + \frac{1}{2})^3$

e. $(a + 3b)^3$

f. $(x^2 + 1)^3$

g. $(x^3 + y)^3$

h. $(5a + 3b)^3$

i. $(x - 3)^3$

j. $(\frac{x}{3} - 4)^3$

k. $(2y - x)^3$

l. $(-x + 3)^3$

m. $(-x - y)^3$

n. $(x - \frac{1}{x})^3$

o. $(x^2 - 1)^3$

p. $(t^3 - 1)^3$

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.5 SUM OF TWO CUBES

Rule

$a^3 + b^3 = (a + b) \cdot (a^2 - ab + b^2)$

Proof

$(a + b) \cdot (a^2 - ab + b^2) = a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3 = a^3 + b^3$

EXAMPLE

Find the products.

a. $(x + 3) \cdot (x^2 - x + 9)$

b. $(x + 1) \cdot (x^2 - x + 1)$

SOLUTION

a. $(x + 3) \cdot (x^2 - x + 9) = (x + 3) \cdot (x^2 - 3x + 3^2) = x^3 + 3^3 = x^3 + 27$

b. $(x + 1) \cdot (x^2 - x + 1) = x^3 + 1^3 = x^3 + 1$

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.6 DIFFERENCE OF TWO CUBES

Rule

$$a^3 - b^3 = (a - b) \cdot (a^2 + ab + b^2)$$

Proof

$$(a - b) \cdot (a^2 + ab + b^2) = a^3 + a^2b + ab^2 - ba^2 - ab^2 + b^3 = a^3 - b^3$$

EXAMPLE

Find the products.

a. $(x - 2y) \cdot (x^2 + 2xy + 4y^2)$

b. $(x - 1) \cdot (x^2 + x + 1)$

SOLUTION

a. $(x - 2y) \cdot (x^2 + 2xy + 4y^2) = (x - 2y) \cdot (x^2 + x \cdot (2y) + (2y)^2)$

$= x^3 - (2y)^3 = x^3 - 8y^3$

b. $(x - 1) \cdot (x^2 + x + 1) = x^3 - 1^3 = x^3 - 1$

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.7 GENERAL EXERCISES

1. Expand the expressions.

a. $(x + 2)^2$ b. $(x - 1)^2$ c. $(2x + 3)^3$

d. $(x - 2y)^3$ e. $(x + 1)^4$ f. $(x - a)^5$

g. $(1 + x)^5$ h. $(a^2 - b^2)^3$ i. $(3x - 1)^6$

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.7 GENERAL EXERCISES

2. Expand the expressions.

a. $(x+\frac{1}{3})^2$ b. $(\sqrt{x}+1)^2$ c. $(\sqrt{x}-\sqrt{y})^2$

d. 98^2 e. 105^2 f. $(x-2y)^2$

g. $(3x^2+2y)^2$ h. $(2xy+1)^2$ i. $(x^2-y^2)^2$

j. $(a+b+1)^2$ k. $(a+2b-c)^2$ l. $(2a-b-3c)^2$

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.7 GENERAL EXERCISES

3. Expand the expressions.

a. $(x + 1)^3$

b. $(x + 2y)^3$

c. $(x - 3)^3$

d. $(x - \frac{1}{x})^3$

e. $(x^2 + 1)^3$

f. $(x^3 - 1)^3$

g. 29^3

h. $(-x + y)^3$

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.7 GENERAL EXERCISES

4. Find the products.

a. $(x + 1) \cdot (x + 1)$

b. $(x + 1) \cdot (x - 1)$

c. $\left(\frac{a}{b} - c\right) \cdot \left(\frac{a}{b} + c\right)$

d. $(\sqrt{x} + \sqrt{y}) \cdot (\sqrt{x} - \sqrt{y})$

e. $(a + b) \cdot (a^2 - ab + b^2)$

f. $(x - y) \cdot (x^2 + xy + y^2)$

g. $(x - 2y) \cdot (x^2 + 2xy + 4y^2)$

h. $(x - 1) \cdot (x^2 + x + 1)$

i. $(x + 3y) \cdot (x^2 - 3xy + 9y^2)$

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.7 GENERAL EXERCISES

5. Calculate $2003^2 - 1997^2$ using an identity.

6. Calculate $2003 \cdot 1997$ using an identity.

2. ALGEBRAIC EXPRESSIONS

2.1 IDENTITIES

2.1.7 GENERAL EXERCISES

7. Calculate the missing term in each sentence.

a. If $x - y = 3$ and $x \cdot y = 2$ then $x^2 + y^2 = ?$

b. If $x + y = 5$ and $x \cdot y = 3$ then $x^2 + y^2 = ?$

c. If $x - y = 7$ and $x \cdot y = 8$ then $x^2 - y^2 = ?$

d. If $x^2 + y^2 = 8$ and $x + y = 3$ then $x \cdot y = ?$

e. If $x - \frac{1}{x} = 5$ then $x^2 + \frac{1}{x^2} = ?$

f. If $x - y = 5$ and $x \cdot y = 3$ then $x^3 - y^3 = ?$

g. If $x^2 + y^2 = 34$ and $x \cdot y = -15$ then $x^3 + y^3 = ?$

2. ALGEBRAIC EXPRESSIONS

2.2 FACTORIZATION

2.2.1 FACTORIZATION METHODS

A. COMMON FACTOR

EXAMPLE

Factor the po

a. $3x^2 - 12x$

b. $4xy^3 - 6x^2y^2$

c. $8a^2b^3 + 12ab^2 - 6a^2b$

$$3x^2 + 15x = (3x) \cdot x + (3x) \cdot 5 = 3x \cdot (x + 5).$$

common monomial factor

2. ALGEBRAIC EXPRESSIONS

2.2 FACTORIZATION

2.2.1 FACTORIZATION METHODS

B. FACTORING BY GROUPING



Factor $ax + by + ay + bx$.



$$ax + by + ay + bx = ax + bx + ay + by$$

$$= x(a + b) + y(a + b)$$

$$= (a + b) \cdot (x + y)$$

2. ALGEBRAIC EXPRESSIONS

2.2 FACTORIZATION

2.2.1 FACTORIZATION METHODS

B. FACTORING BY GROUPING

EXAMPLE

Factor the polynomials.

a. $x^3 + x^2 + 4x + 4$

b. $ax - 3a - bx + 3b$

c. $xy + 2x - y - 2$

d. $x^2 + yz + xy + xz$

e. $a^3 + a^2 + a + 1$

EXAMPLE

Factor $x^2 + 2xy + y^2 - z^2$.

EXAMPLE

Factor the polynomials.

a. $x^2 - 1$

b. $x^2 - 4$

c. $x^2 - 9$

d. $4 - x^2y^2$

e. $81 \cdot a^{12} - 16 \cdot a^8$

EXAMPLE

Factor $4 - 4xy - x^2 - 4y^2$.

2. ALGEBRAIC EXPRESSIONS

2.2 FACTORIZATION

2.2.1 FACTORIZATION METHODS

C. FACTORING PERFECT SQUARE TRINOMIALS

Recall the expansions of $(a + b)^2$ and $(a - b)^2$:

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2.$$

$(a + b)^2$ and $(a - b)^2$ are perfect squares.

Their equivalent trinomials $a^2 + 2ab + b^2$ and $a^2 - 2ab + b^2$ are called **perfect square trinomials**.

2. ALGEBRAIC EXPRESSIONS

2.2 FACTORIZATION

2.2.1 FACTORIZATION METHODS

C. FACTORING PERFECT SQUARE TRINOMIALS

EXAMPLE

Factor the polynomials.

a. $x^2 + 2x + 1$

b. $x^2 + 4x + 4$

c. $x^2 - 10x + 25$ ⟶

d. $16x^2 + 24x + 9$ ⟶

e. $4y^2 - 20y + 25$ ⟶

SOLUTION

a. $x^2 + 2x + 1 = x^2 + 2 \cdot x \cdot 1 + 1^2 = (x + 1)^2$

b. $x^2 + 4x + 4 = x^2 + 2 \cdot x \cdot 2 + 2^2 = (x + 2)^2$

2. ALGEBRAIC EXPRESSIONS

2.2 FACTORIZATION

2.2.1 FACTORIZATION METHODS

D. COMPLETING THE SQUARE



Factorize $x^2 + 5x + 4$.

2. ALGEBRAIC EXPRESSIONS

2.2 FACTORIZATION

2.2.1 FACTORIZATION METHODS

D. COMPLETING THE SQUARE

EXAMPLE

Factorize $x^2 - x - 1$.

2. ALGEBRAIC EXPRESSIONS

2.2 FACTORIZATION

2.2.1 FACTORIZATION METHODS

E. OTHER FACTORING METHODS

a. The form $x^2 + bx + c$

Now let $x^2 + bx + c$ be a monic polynomial.

We can factorize this polynomial into two binomials by using the following procedure:

First identify two factors m and n of c whose sum is equal to b, so $c = m \cdot n$ and $b = m + n$.

Then $x^2 + bx + c = x^2 + (m + n)x + m \cdot n$

$= x^2 + mx + nx + m \cdot n$

$= x \cdot (x + m) + n \cdot (x + m)$

$= (x + m) \cdot (x + n)$.

In short, $x^2 + bx + c = (x + m) \cdot (x + n)$.

$m + n$ $m \cdot n$

For example, let us factorize $x^2 + 5x - 6$ using this method:

$x^2 + 5x - 6 = (x + 6) \cdot (x - 1)$

$6+(-1)$ $6 \cdot (-1)$

8th-Grade

2. ALGEBRAIC EXPRESSIONS

2.2 FACTORIZATION

2.2.1 FACTORIZATION METHODS

E. OTHER FACTORING METHODS

a. The form $x^2 + bx + c$

EXAMPLE

Factorize the polynomials.

a. $x^2 + 8x + 7$

b. $x^2 + 7x + 10$

c. $x^2 - 4x - 12$

d. $x^2 - x - 2$

e. $x^2 + 7x - 18$

f. $x^2 - 7x + 10$

SOLUTION

a. $x^2 + 8x + 7 = (x + 7) \cdot (x + 1)$

$7 + 1 \quad 7 \cdot 1$

8th-Grade

2. ALGEBRAIC EXPRESSIONS

2.2 FACTORIZATION

2.2.1 FACTORIZATION METHODS

E. OTHER FACTORING METHODS

a. The form $x^2 + bx + c$

PRACTICE PROBLEMS

Factorize the polynomials.

a. $x^2 - 2x - 3$

b. $x^2 - 8x - 9$

c. $x^2 - x - 56$

d. $x^2 + 5x - 36$

e. $x^2 - 7x - 18$

f. $x^2 - 7x + 12$

g. $y^2 - 5y + 6$

h. $t^2 + t - 110$

i. $m^2 - m \cdot a - 6a^2$

j. $y^2 - y - 20$

2. ALGEBRAIC EXPRESSIONS

2.2 FACTORIZATION

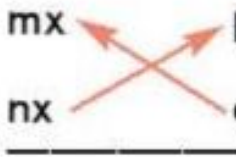
2.2.1 FACTORIZATION METHODS

E. OTHER FACTORING METHODS

b. The form $ax^2 + bx + c$

Rule

$$ax^2 + bx + c = mnx^2 + (mq+np)x + p \cdot q = (mx+p) \cdot (nx+q)$$



$$mqx + npx = (mq + np)x$$

where m and n are factors of a and p and q are factors of c.

2. ALGEBRAIC EXPRESSIONS

2.2 FACTORIZATION

2.2.1 FACTORIZATION METHODS

E. OTHER FACTORING METHODS

b. The form $ax^2 + bx + c$

EXAMPLE

Factorize $4x^2 - 7x - 2$.

EXAMPLE

Factorize $2x^2 + 13x + 21$.

2. ALGEBRAIC EXPRESSIONS

2.2 FACTORIZATION

2.2.1 FACTORIZATION METHODS

E. OTHER FACTORING METHODS

b. The form $ax^2 + bx + c$

PRACTICE PROBLEMS

Factor the polynomials.

a. $6x^2 + 5x - 4$

b. $3x^2 + 15x + 12$

c. $6x^2 + 13x + 6$

d. $4x^2 + 11x - 3$

e. $2x^2 - 5x + 3$

f. $4y^2 - 13y + 10$

g. $2(x + 3)^2 - 7 \cdot (x + 3) - 4$

h. $4x^3 - 5x^2 + x$

i. $36x^2 + 30x - 50$

j. $2(x - y)^2 + 5(x - y) + 3$

k. $x^4 + 12x^2 + 32$

l. $x^{2y} - 5x^y - 24$

2. ALGEBRAIC EXPRESSIONS

2.3 CHAPTER REVIEW EXERCISES

1. Find all the factors of each number.

a. 18 b. 48 c. 60 d. 96 e. 128

2. ALGEBRAIC EXPRESSIONS

2.3 CHAPTER REVIEW EXERCISES

2. Find the greatest common factor of each pair of numbers.

a. 12, 27

b. 24, 56

c. 60, 72

d. 36, 48, 60

2. ALGEBRAIC EXPRESSIONS

2.3 CHAPTER REVIEW EXERCISES

3. Find the greatest common factor of the given monomials.

a. $3x, 6xy$

b. $8xy^2, 12x^2y^2$

c. $15x^2yz^2, 20x^3y^2z$

d. $3x^2yz, 9xy^2z, 27xyz^2$

2. ALGEBRAIC EXPRESSIONS

2.3 CHAPTER REVIEW EXERCISES

4. Factor the polynomials.

a. $ab + ac$

b. $xy + 3x^2y - 2xy^2$

c. $35x^2y^3 + 21xy^4$

d. $13a^2b^2c^2 + 26a^3bc^2 - 52a^3bc$

2. ALGEBRAIC EXPRESSIONS

2.3 CHAPTER REVIEW EXERCISES

5. Factor by grouping.

a. $ax + by + ay + bx$

b. $ma - na + mb - nb$

c. $a^3 - a^2 + ab - b$

d. $z^2 + 2z + 3zy + 6y$

e. $x^2 - 3x - xy + 3y$

f. $a^3 + a^2b - a - b$

g. $6a^2 + 2a - 3ab - b$

h. $x^2 - y^2 - x - y$

i. $a - 2b + a^2 - 4b^2$

2. ALGEBRAIC EXPRESSIONS

2.3 CHAPTER REVIEW EXERCISES

6. Factor by using the difference of two squares.

a. $9a^2 - 64$

b. $81x^4 - 16y^2$

c. $64x^4 - 144y^6$

d. $(a + b)^2 - c^2$

e. $a^2 - (b - c)^2$

f. $7x^2 - 112$

g. $5x^3 - 180x$

h. $a^{2x} - b^{2y}$

i. $a^{6m} - b^{6n}$

2. ALGEBRAIC EXPRESSIONS

2.3 CHAPTER REVIEW EXERCISES

7. Factor by using the difference of two cubes.

a. $(2x)^3 - 8$

b. $a^3 - 64b^3$

c. $3x^3 - 81y^3z^6$

d. $16a^3b^3 - 54c^3d^3$

2. ALGEBRAIC EXPRESSIONS

2.3 CHAPTER REVIEW EXERCISES

8. Factor.

a. $x^5 - (2y)^5$

b. $a^5 + b^5$

c. $1 + m^5$

d. $125 - x^6$

2. ALGEBRAIC EXPRESSIONS

2.3 CHAPTER REVIEW EXERCISES

9. Factor the trinomials.

a. $x^2 + 3x - 4$

b. $x^2 + 2x - 15$

c. $x^2 - 9x + 20$

d. $x^2 - 4x - 21$

e. $x^2 - 6x - 27$

f. $x^2 - 10x + 9$

2. ALGEBRAIC EXPRESSIONS

2.3 CHAPTER REVIEW EXERCISES

10. Factor the trinomials.

a. $2x^2 + 5x - 3$

b. $3x^2 + 10x + 3$

c. $3x^2 - 7x - 6$

d. $6x^2 + 7x - 3$

e. $8x^2 + 42x + 49$

f. $14x^2 + 47x - 27$

2. ALGEBRAIC EXPRESSIONS

2.3 CHAPTER REVIEW EXERCISES

11. Factor each expression and simplify it.

a. $4 \cdot (x-1)^2 - 6 \cdot (x-1) - 10$

b. $6x^{2y} - 7x^{y} - 5$

2. ALGEBRAIC EXPRESSIONS

2.3 CHAPTER REVIEW EXERCISES

12. $8x^2 + (3a - 1)x + 45 = (4x - 5) \cdot (2x - 9)$ is given.
What is the value of a?